

# Risk Intelligence Report

CFASimplified Agentic Pipeline

Domain: Quantum Computing Cryptography Risk

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Quick Facts:

|                         |                                                |
|-------------------------|------------------------------------------------|
| Total Records Collected | 1287                                           |
| Sources Enabled         | NEWS, FINANCIAL, WEATHER, SOCIAL               |
| Primary Keywords        | quantum computing, cryptography, cybersecurity |

# ■ Data Collection Summary

**Total Records Collected:** 1287  
**Collection Completeness:** All available sources processed  
**Keywords Used:** quantum computing, cryptography, cybersecurity, encryption, data security

## Records by Source:

| Source      | Records | Percentage |
|-------------|---------|------------|
| SOCIAL      | 545     | 42.3%      |
| COMMODITIES | 478     | 37.1%      |
| NEWS        | 260     | 20.2%      |
| WEATHER     | 4       | 0.3%       |
| STOCKS      | 0       | 0.0%       |
| TOTAL       | 1287    | 100.0%     |

# ■■ Top 5 Identified Risks

**Risk #1: Post-Quantum Key Exchange Risks**  
**Category:** Quantum Cryptography Vulnerability  
**Severity:** ■ HIGH  
**Confidence:** 80%  
**Probability (30 days):** 60%

**Key Evidence:**

- Red Hat Enterprise Linux post-quantum key exchange algorithms
- Importance of randomness in encryption

**Recommended Action:** Implement post-quantum key exchange algorithms

**Risk #2: Misunderstanding Quantum Computing**  
**Category:** Quantum Computing Misinformation  
**Severity:** ■ MEDIUM  
**Confidence:** 70%  
**Probability (30 days):** 50%

**Key Evidence:**

- Negative social media posts about quantum computing explanations
- Lack of understanding of quantum computing principles

**Recommended Action: Improve quantum computing education and awareness**

**Risk #3: Randomness in Encryption**

**Category:** Cryptography Security

**Severity:** ■ MEDIUM

**Confidence:** 60%

**Probability (30 days):** 40%

**Key Evidence:**

- Importance of randomness in encryption
- Quantum computing's potential to generate perfect randomness

**Recommended Action: Explore quantum computing-based randomness generation for encryption**

**Risk #4: Slow Adoption of Quantum Computing**

**Category:** Quantum Computing Adoption

**Severity:** ■ LOW

**Confidence:** 50%

**Probability (30 days):** 30%

**Key Evidence:**

- Limited social media posts about quantum computing
- Lack of mainstream quantum computing applications

**Recommended Action: Increase quantum computing awareness and education**

**Risk #5: Limited Quantum Computing Research**

**Category:** Quantum Computing Research

**Severity:** ■ LOW

**Confidence:** 40%

**Probability (30 days):** 20%

**Key Evidence:**

- Limited research papers on quantum computing
- Few social media posts about quantum computing research

**Recommended Action: Increase funding and support for quantum computing research**

## ■ Counterfactual Analysis & Interventions

This section shows **what-if** scenarios where we test interventions to reduce risk.

### Risk #1: Post-Quantum Key Exchange Risks

**1. Scenario ID:** cf\_1\_b\_sup

**Intervention Type:** SUPPLY

**Strategy:** Activate a dual/multi-sourcing programme for critical inputs linked to 'Post-Quantum Key Exchange Risks': qualify at least two geographically diverse suppliers within 60 days and pre-negotiate surge-capacity agreements.

**Baseline Risk Probability:** 60%

**After Intervention:** 41% (↓ 19%)

**Implementation Feasibility:** HIGH

**Estimated Cost:** \$500K – \$10M (inventory / sourcing)

**Time to Impact:** 14 days

**2. Scenario ID:** cf\_1\_a\_pol

**Intervention Type:** POLICY

**Strategy:** Implement a formal quantum\_computing\_cryptography\_risk risk-governance policy that addresses 'Post-Quantum Key Exchange Risks': mandate cross-functional oversight, set exposure limits, and establish escalation protocols aligned with 'Implement post-quantum key exchange algorithms'.

**Baseline Risk Probability:** 60%

**After Intervention:** 47% (↓ 13%)

**Implementation Feasibility:** HIGH

**Estimated Cost:** \$50K – \$500K (staffing + legal)

**Time to Impact:** 60 days

## **Risk #2: Misunderstanding Quantum Computing**

**1. Scenario ID:** cf\_2\_b\_sup

**Intervention Type:** SUPPLY

**Strategy:** Activate a dual/multi-sourcing programme for critical inputs linked to 'Misunderstanding Quantum Computing': qualify at least two geographically diverse suppliers within 60 days and pre-negotiate surge-capacity agreements.

**Baseline Risk Probability:** 50%

**After Intervention:** 35% (↓ 15%)

**Implementation Feasibility:** HIGH

**Estimated Cost:** \$500K – \$10M (inventory / sourcing)

**Time to Impact:** 14 days

**2. Scenario ID:** cf\_2\_a\_pol

**Intervention Type:** POLICY

**Strategy:** Implement a formal quantum\_computing\_cryptography\_risk risk-governance policy that addresses 'Misunderstanding Quantum Computing': mandate cross-functional oversight, set exposure limits, and establish escalation protocols aligned with 'Improve quantum computing education and awareness'.

**Baseline Risk Probability:** 50%

**After Intervention:** 40% (↓ 10%)

**Implementation Feasibility:** HIGH

**Estimated Cost:** \$50K – \$500K (staffing + legal)

**Time to Impact:** 60 days

### **Risk #3: Randomness in Encryption**

**1. Scenario ID:** cf\_3\_b\_sup

**Intervention Type:** SUPPLY

**Strategy:** Activate a dual/multi-sourcing programme for critical inputs linked to 'Randomness in Encryption': qualify at least two geographically diverse suppliers within 60 days and pre-negotiate surge-capacity agreements.

**Baseline Risk Probability:** 40%

**After Intervention:** 29% (↓ 11%)

**Implementation Feasibility:** HIGH

**Estimated Cost:** \$500K – \$10M (inventory / sourcing)

**Time to Impact:** 14 days

**2. Scenario ID:** cf\_3\_a\_pol

**Intervention Type:** POLICY

**Strategy:** Implement a formal quantum\_computing\_cryptography\_risk risk-governance policy that addresses 'Randomness in Encryption': mandate cross-functional oversight, set exposure limits, and establish escalation protocols aligned with 'Explore quantum computing-based randomness generation for encryption'.

**Baseline Risk Probability:** 40%

**After Intervention:** 32% (↓ 8%)

**Implementation Feasibility:** HIGH

**Estimated Cost:** \$50K – \$500K (staffing + legal)

**Time to Impact:** 60 days

## **Risk #4: Slow Adoption of Quantum Computing**

**1. Scenario ID:** cf\_4\_b\_sup

**Intervention Type:** SUPPLY

**Strategy:** Activate a dual/multi-sourcing programme for critical inputs linked to 'Slow Adoption of Quantum Computing': qualify at least two geographically diverse suppliers within 60 days and pre-negotiate surge-capacity agreements.

**Baseline Risk Probability:** 30%

**After Intervention:** 23% (↓ 7%)

**Implementation Feasibility:** HIGH

**Estimated Cost:** \$500K – \$10M (inventory / sourcing)

**Time to Impact:** 14 days

**2. Scenario ID:** cf\_4\_a\_pol

**Intervention Type:** POLICY

**Strategy:** Implement a formal quantum\_computing\_cryptography\_risk risk-governance policy that addresses 'Slow Adoption of Quantum Computing': mandate cross-functional oversight, set exposure limits, and establish escalation protocols aligned with 'Increase quantum computing awareness and education'.

**Baseline Risk Probability:** 30%

**After Intervention:** 25% (↓ 5%)

**Implementation Feasibility:** HIGH

**Estimated Cost:** \$50K – \$500K (staffing + legal)

**Time to Impact:** 60 days

## **Risk #5: Limited Quantum Computing Research**

**1. Scenario ID:** cf\_5\_a\_sup

**Intervention Type:** SUPPLY

**Strategy:** Activate a dual/multi-sourcing programme for critical inputs linked to 'Limited Quantum Computing Research': qualify at least two geographically diverse suppliers within 60 days and pre-negotiate surge-capacity agreements.

**Baseline Risk Probability:** 20%

**After Intervention:** 16% (↓ 4%)

**Implementation Feasibility:** HIGH

**Estimated Cost:** \$500K – \$10M (inventory / sourcing)

**Time to Impact:** 14 days

**2. Scenario ID:** cf\_5\_b\_pol

**Intervention Type:** POLICY

**Strategy:** Implement a formal quantum\_computing\_cryptography\_risk risk-governance policy that addresses 'Limited Quantum Computing Research': mandate cross-functional oversight, set exposure limits, and establish escalation protocols aligned with 'Increase funding and support for quantum computing research'.

**Baseline Risk Probability:** 20%

**After Intervention:** 17% (↓ 3%)

**Implementation Feasibility:** HIGH

**Estimated Cost:** \$50K – \$500K (staffing + legal)

**Time to Impact:** 60 days



## ■ Recommended Solutions & Mitigations

Based on the risk analysis and counterfactual interventions, here are the top recommended solutions:

### Risk: Post-Quantum Key Exchange Risks

#### 1. Post-Quantum Key Exchange Implementation [TECHNOLOGY]

**Type:** TECHNOLOGY | **Time Horizon:** MEDIUM

**Relevance Score:** 40.0% | **Risk Reduction:** 20-30% probability reduction

**Estimated Cost:** 100,000 - 500,000

**Description:** Implementing post-quantum key exchange algorithms is crucial to mitigate the risk of classical encryption being cracked by quantum computers. Although the provided KB chunks do not directly address post-quantum cryptography, they emphasize the importance of adopting policies and frameworks for risk governance and management. Thus, a technology-based solution focusing on post-quantum key exchange a

##### Key Implementation Steps:

- Conduct a thorough review of existing cryptographic systems to identify areas vulnerable to quantum computing attacks
- Research and select suitable post-quantum key exchange algorithms (e.g., lattice-based, code-based, or hash-based cryptography) for implementation
- Develop and deploy the chosen post-quantum key exchange algorithms, ensuring backward compatibility with existing systems where necessary

#### 2. Dual Sourcing Framework [FRAMEWORK]

**Type:** FRAMEWORK | **Time Horizon:** MEDIUM

**Relevance Score:** 40.0% | **Risk Reduction:** 10-20% probability reduction

**Estimated Cost:** 100k-500k

**Description:** Implementing a dual sourcing framework can reduce the risk of post-quantum key exchange by qualifying at least two geographically diverse suppliers per critical component. This approach can reduce single-point-of-failure exposure by 40-60%. However, the provided KB chunks have limited direct relevance to post-quantum key exchange risks, so the relevance score is lower. A dual sourcing framework ca

##### Key Implementation Steps:

- Conduct a thorough analysis of the supply chain for post-quantum key exchange components
- Identify and qualify at least two geographically diverse suppliers per critical component
- Implement annual supplier scorecards covering quality, delivery, financial health, and geopolitical risk

## **Risk: Misunderstanding Quantum Computing**

### **1. Quantum Computing Awareness Campaign [PROCESS]**

**Type:** PROCESS | **Time Horizon:** SHORT

**Relevance Score:** 40.0% | **Risk Reduction:** 10-20% probability reduction

**Estimated Cost:** 100,000 - 500,000

**Description:** This solution fits the risk of misunderstanding quantum computing by implementing a process to improve education and awareness. Although the provided KB chunks do not directly address quantum computing, a similar approach to the Digital Misinformation Rapid Response System can be applied to create a social listening dashboard and counter-narratives for quantum computing misinformation. The relevant

#### **Key Implementation Steps:**

- Develop a social listening dashboard to detect quantum computing misinformation trends
- Create a library of evidence-based counter-narratives for quantum computing
- Partner with social media platforms for coordinated takedowns of misinformation

### **2. Quantum Education Partnership [PARTNERSHIP]**

**Type:** PARTNERSHIP | **Time Horizon:** LONG

**Relevance Score:** 40.0% | **Risk Reduction:** 30-50% probability reduction

**Estimated Cost:** 10M-50M

**Description:** This solution involves forming partnerships between academic institutions, industry leaders, and government agencies to develop and implement quantum computing education and awareness programs. The goal is to improve the general public's understanding of quantum computing and its potential applications. Although the provided KB chunks do not directly address quantum computing education, they highlight

#### **Key Implementation Steps:**

- Establish a steering committee comprising representatives from academia, industry, and government to oversee the development of quantum computing education programs
- Develop a comprehensive curriculum for quantum computing education, incorporating input from industry experts and academic researchers
- Implement a nationwide outreach and awareness campaign to promote the importance of quantum computing education and its potential applications

## Risk: Randomness in Encryption

### 1. Quantum Randomness Generation [TECHNOLOGY]

**Type:** TECHNOLOGY | **Time Horizon:** SHORT

**Relevance Score:** 30.0% | **Risk Reduction:** 10-20% probability reduction

**Estimated Cost:** 100,000 - 500,000

**Description:** Although the provided KB chunks do not directly address quantum computing or cryptography, a potential short-term mitigation could involve exploring the application of AI or automation principles to enhance encryption security. However, due to the weak coverage of the topic in the KB, the relevance score is lowered. This solution focuses on the potential for technology to improve randomness in enc

#### Key Implementation Steps:

- Research existing technologies for quantum randomness generation
- Assess the feasibility of integrating such technologies into current encryption systems
- Develop a pilot project to test the effectiveness of quantum randomness generation in improving encryption security

### 2. Strategic Supply Chain Framework [FRAMEWORK]

**Type:** FRAMEWORK | **Time Horizon:** MEDIUM

**Relevance Score:** 42.0% | **Risk Reduction:** 15-30% probability reduction

**Estimated Cost:** 100k-500k

**Description:** This solution fits the risk of randomness in encryption by implementing a strategic supply chain framework that reduces single-point-of-failure exposure. Although the provided KB chunks do not directly address quantum computing-based randomness generation, a dual/multi-sourcing strategy can be applied to critical inputs linked to encryption, enhancing overall supply chain resilience. The relevance

#### Key Implementation Steps:

- Conduct a thorough analysis of the supply chain for critical encryption components
- Identify and qualify at least two geographically diverse suppliers per critical component
- Implement annual supplier scorecards covering quality, delivery, financial health, and geopolitical risk

## ■ Report Summary

This Risk Intelligence Report synthesizes multi-source signals across **NEWS, FINANCIAL, WEATHER, SOCIAL** to identify, analyze, and recommend mitigations for risks in the **Quantum Computing Cryptography Risk** domain.

#### Pipeline Overview:

**Layer 1:** Collected 1287 signals from multiple sources

**Layer 2:** Enriched signals with NLP sentiment, entities, and reliability scoring

**Layer 3:** Identified and ranked top 5 risks using LLM analysis

**Layer 4:** Tested counterfactual interventions to find risk-reducing actions

**Layer 5:** Mapped interventions to practical solutions from knowledge base

#### Next Steps:

1. Review the top identified risks and prioritize by severity and probability
2. Evaluate counterfactual interventions for feasibility and cost-benefit
3. Implement recommended solutions in order of priority and time-to-impact

#### 4. Monitor KPIs and re-run analysis as conditions evolve

**Report Generated By:** CFASimplified Agentic AI Pipeline

**Timestamp:** 2026-05-29 22:20:30

**Disclaimer:** This report is for informational purposes and should be reviewed by domain experts before decision-making.